

Software Engineering Lab -1

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Section: E

1) Functional Requirements:

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse inbound Shopify/WooCommerce order webhooks and create an internal order record containing the customer and product details.	High	Pass: A valid webhook creates an internal order with all required order details. Fail: An invalid or incomplete webhook creates an incomplete order.	Orders must be captured accurately before further processing.
FR-002	Functional	The system shall verify real-time supplier inventory availability for every SKU included in an order.	High	Pass: The system retrieves the current supplier stock status for each SKU. Fail: A purchase order is generated for an unavailable SKU.	Prevents purchase orders from being sent for out-of-stock products.
FR-003	Functional	The system shall split an order into separate supplier-specific orders when the order contains products supplied by multiple suppliers.	High	Pass: Each supplier receives an order containing only its applicable products. Fail: Products belonging to different suppliers are incorrectly combined into one supplier order.	Multi-vendor orders must be correctly distributed among suppliers.

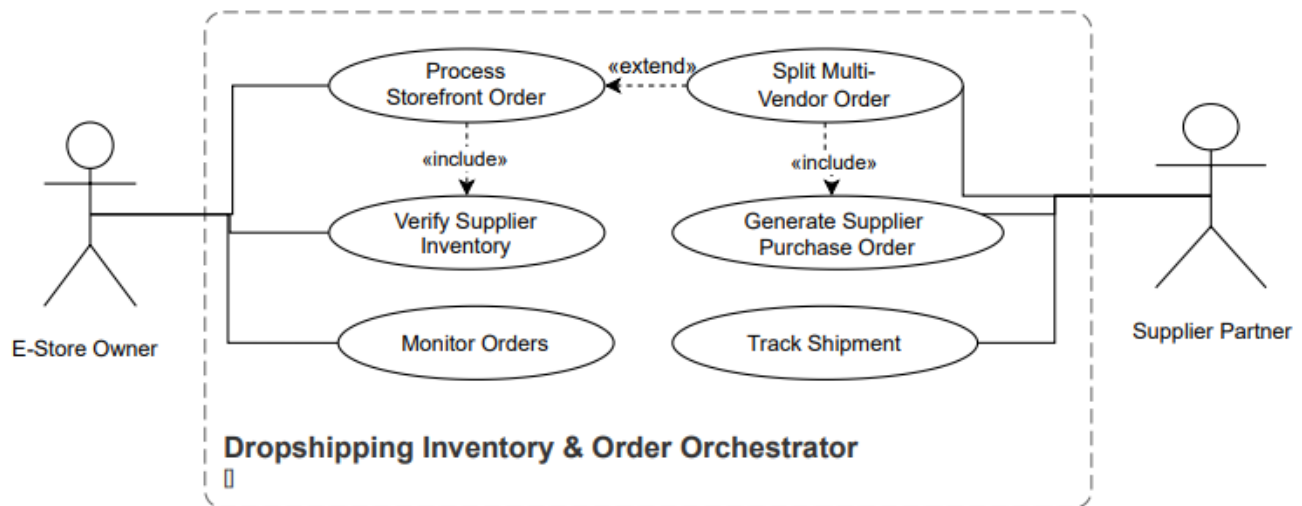
FR-004	Functional	The system shall automatically generate and dispatch a supplier purchase order for each order or order segment with available inventory.	High	Pass: A supplier purchase order is generated and dispatched with the required product, quantity, and customer delivery details. Fail: No purchase order is generated or required information is missing.	Automates the supplier ordering process and reduces manual intervention.
FR-005	Functional	The system shall record and update shipment tracking numbers received from supplier partners and associate them with the corresponding customer order.	Medium	Pass: A valid tracking number is stored against the correct order and can be retrieved. Fail: The tracking number is associated with the wrong order or cannot be retrieved.	Enables the store owner to monitor supplier shipments and customer orders.

Non Functional Requirements:

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	Inbound webhook processing shall acknowledge receipt with HTTP 200 within 200 ms under normal operating conditions.	High	Pass: Benchmarking confirms the acknowledgement target is met under simulated peak load while maintaining required security controls. Fail: The response exceeds 200 ms or security requirements are violated.	Fast acknowledgement prevents webhook retry storms and improves reliability.

NFR-002	Reliability	The system shall process each valid storefront order webhook without creating duplicate supplier purchase orders when the same webhook is delivered more than once.	High	Pass: Repeated delivery of the same webhook results in one logical order and one corresponding supplier PO per supplier. Fail: Duplicate purchase orders are generated.	Webhook providers may retry requests, so duplicate processing could cause duplicate supplier orders.
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2)UML Diagram



3) Use-Case Flow Specification

Process Storefront Order

Primary Actor: E-Store Owner

Supporting Actor: Supplier Partner

Preconditions

1. The e-commerce storefront is connected to the Dropshipping Inventory & Order Orchestrator.
2. A valid customer order webhook is received from the storefront.
3. Supplier inventory information is available for the products in the order.

Postconditions

Success:

- The customer order is recorded in the system.
- Supplier inventory has been verified.
- The order is split into supplier-specific segments when multiple suppliers are involved.
- The required supplier purchase orders are generated and dispatched.
- The order is available for shipment tracking.

Failure:

- A purchase order is not generated for an unavailable supplier SKU.
- The affected order or order segment is flagged for further attention.

Main Success Scenario

1. The storefront sends an order webhook to the system.
2. The system acknowledges and parses the incoming order information.
3. The system creates an internal order record containing the required order and product details.
4. The system identifies the supplier associated with each SKU.
5. The system verifies the current inventory availability with the relevant supplier.
6. The supplier confirms that the required products are available.
7. The system determines whether the order contains products from multiple suppliers.
8. If multiple suppliers are involved, the system separates the order into supplier-specific segments.
9. The system generates a supplier purchase order for each applicable supplier.
10. The system dispatches the purchase orders to the suppliers.
11. The system records the purchase-order status.

12. When shipment information is received, the system records the tracking number against the corresponding customer order.
13. The order is made available for monitoring and shipment tracking by the E-Store Owner.

Alternate Flow:

At Step 6: if the supplier reports that a required SKU is unavailable:

1. The system marks the SKU as unavailable.
2. The system does not generate a purchase order for the unavailable SKU.
3. The system records the affected order segment as requiring attention.
4. The affected order is flagged for the E-Store Owner.
5. If other order segments have available inventory, those segments continue through the normal purchase-order process.